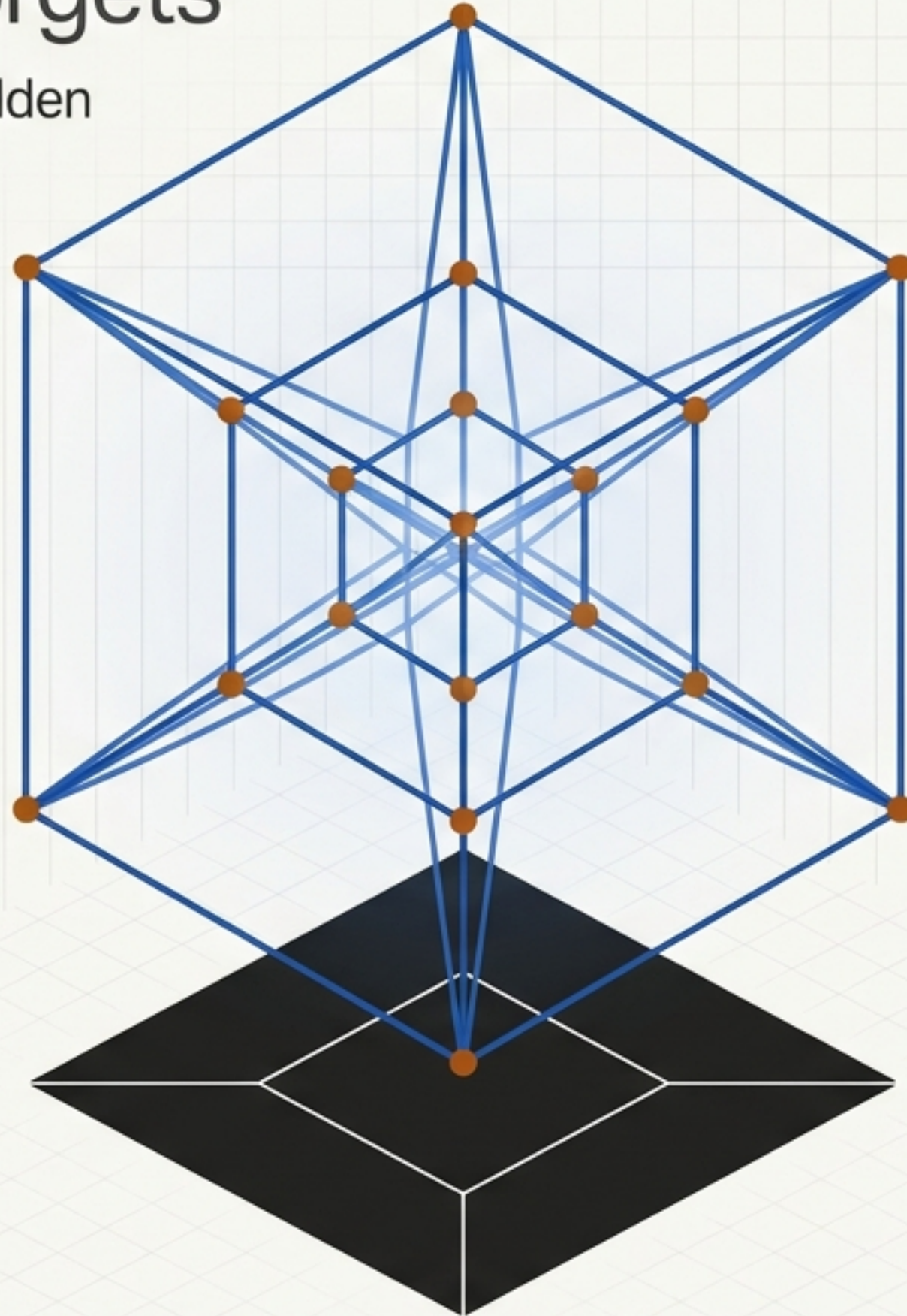


What Probability Forgets

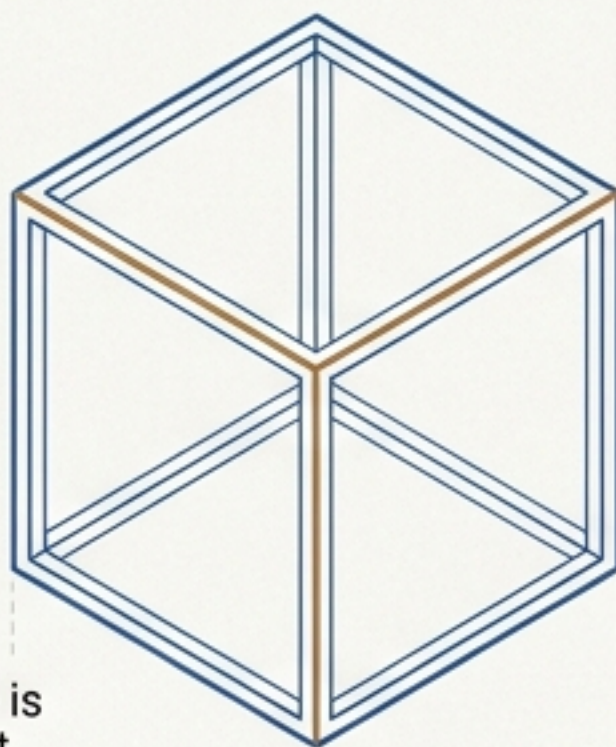
Projection, Interference, and the Hidden
Architecture of Quantum Theory

Flyxion

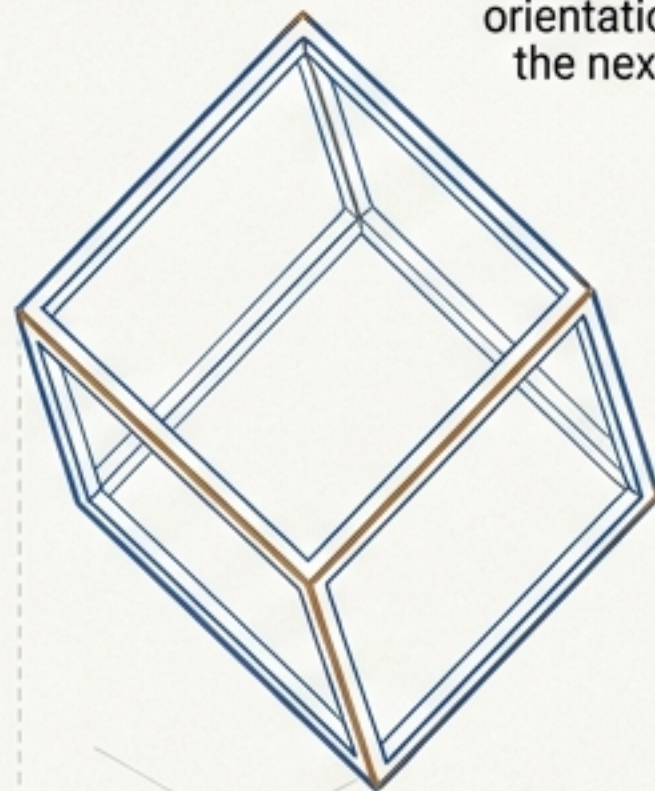


The Honest Shadow

A probability table is perfectly honest. It accurately records a single observation.

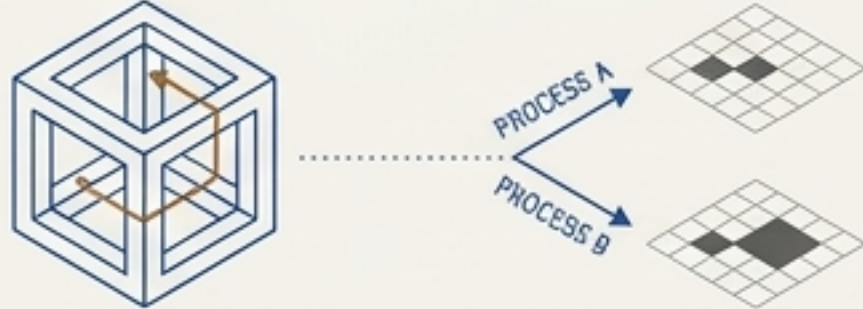
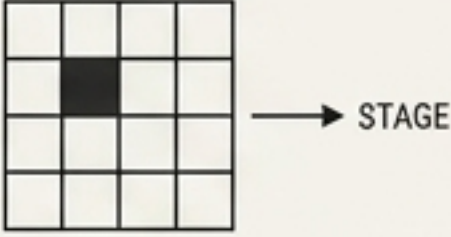
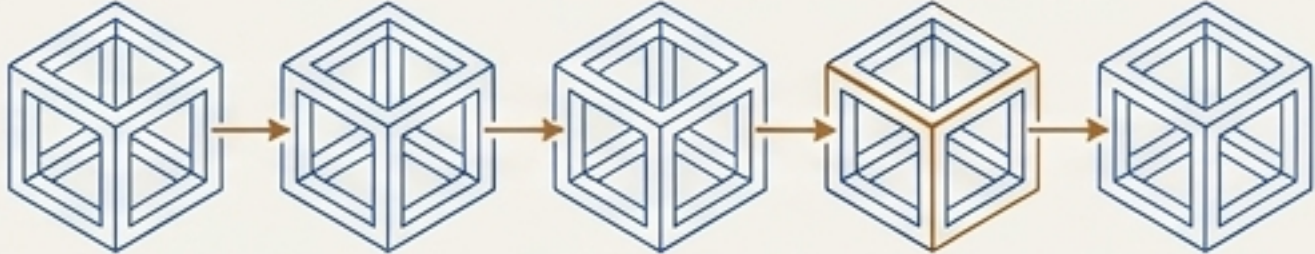
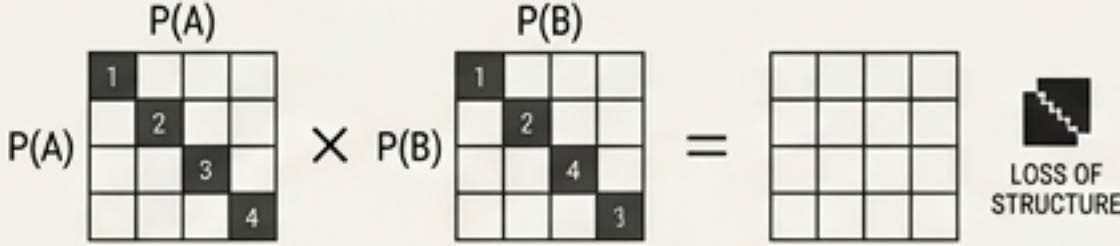
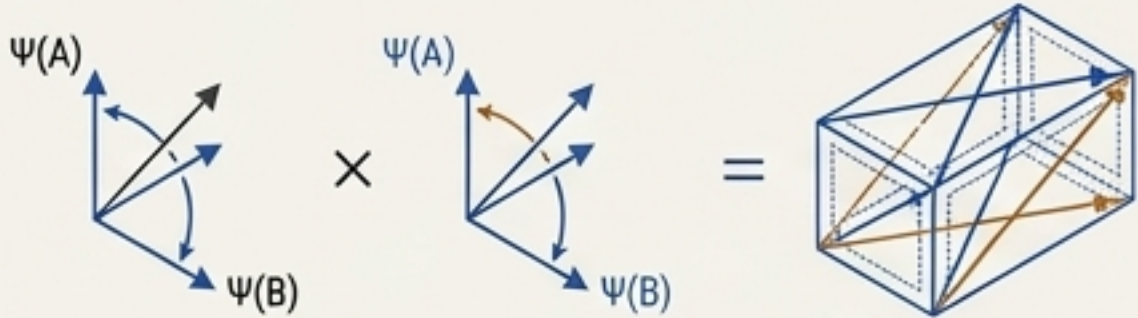


But projection erases depth. It forgets the relational orientation required for the next movement.



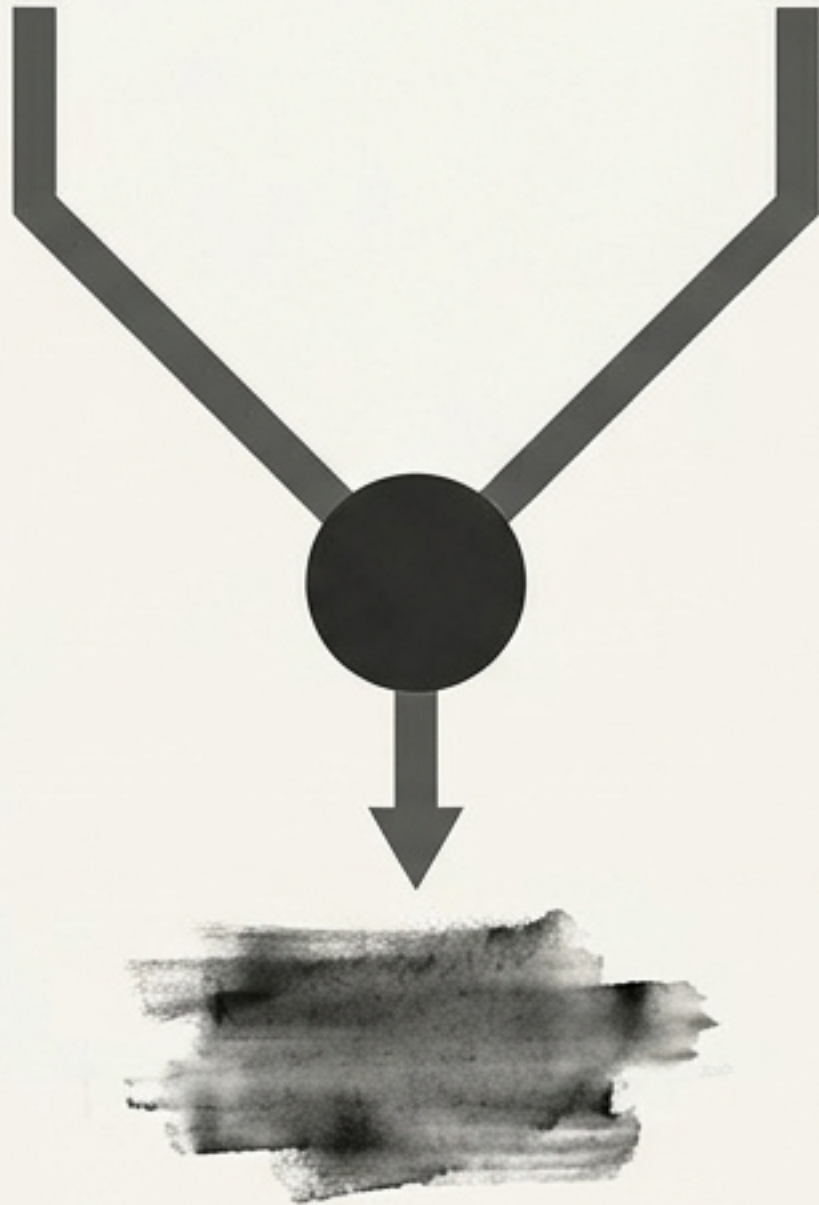
The missing information remains harmless until we ask the system to continue. Then, the omission returns as a surprise.

Two Levels of Description

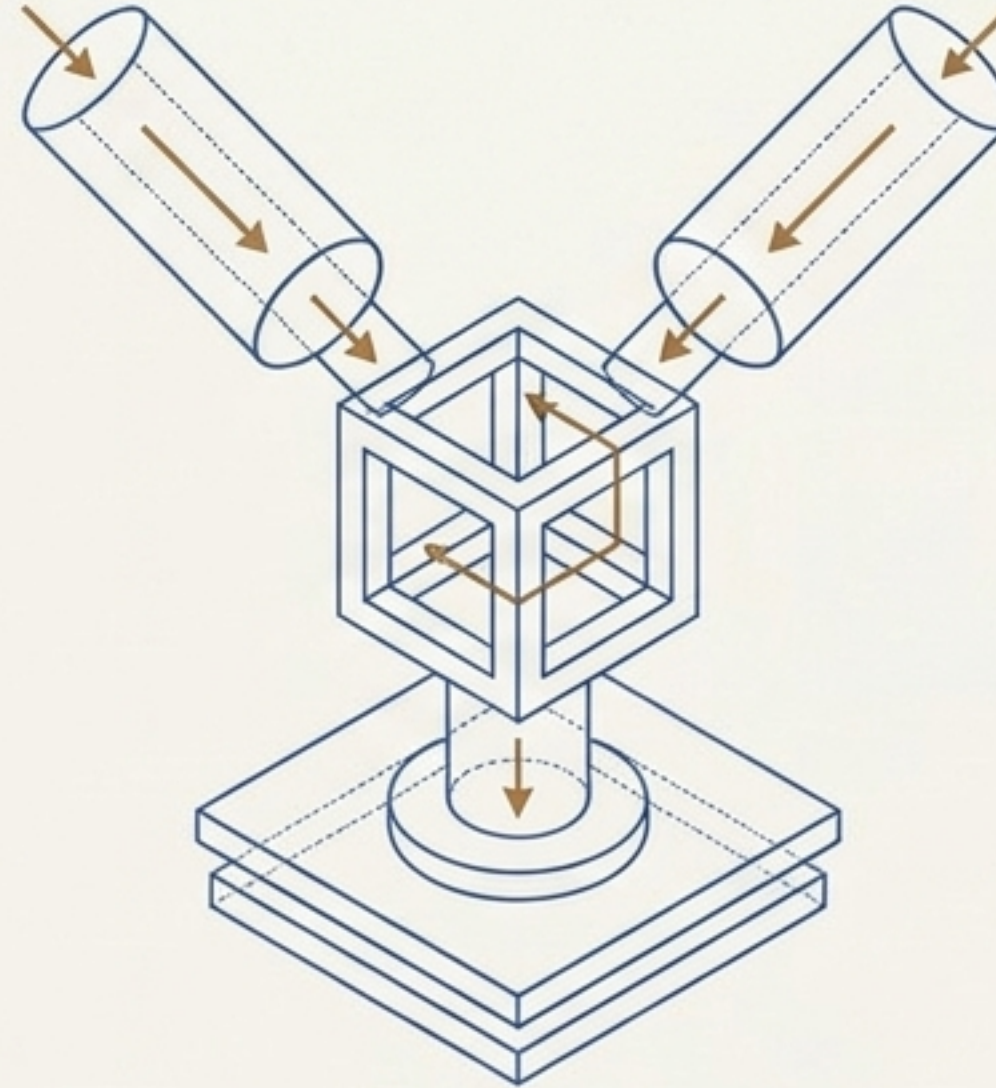
	The Witness (Probability)	The Mechanism (Amplitude)
Function	 Records an outcome and summarizes visible consequences.	 Supports counterfactuals and distinguishes processes casting the same shadow.
Sufficiency	 Sufficient for one isolated experimental stage.	 Sufficient for coherent continuation and composition.
Result of Composition	 Multiply probabilities = Loss of relational structure.	 Multiply amplitudes = Phase terms dictate reinforcement or cancellation.

"Sufficiency is relative to the task. A probability table behaves exactly as designed; we simply ask it to retain information it was never built to carry."

Interference as Returning Memory



Forget, then Compose. Cross-terms are destroyed. Result is a fully mixed table.

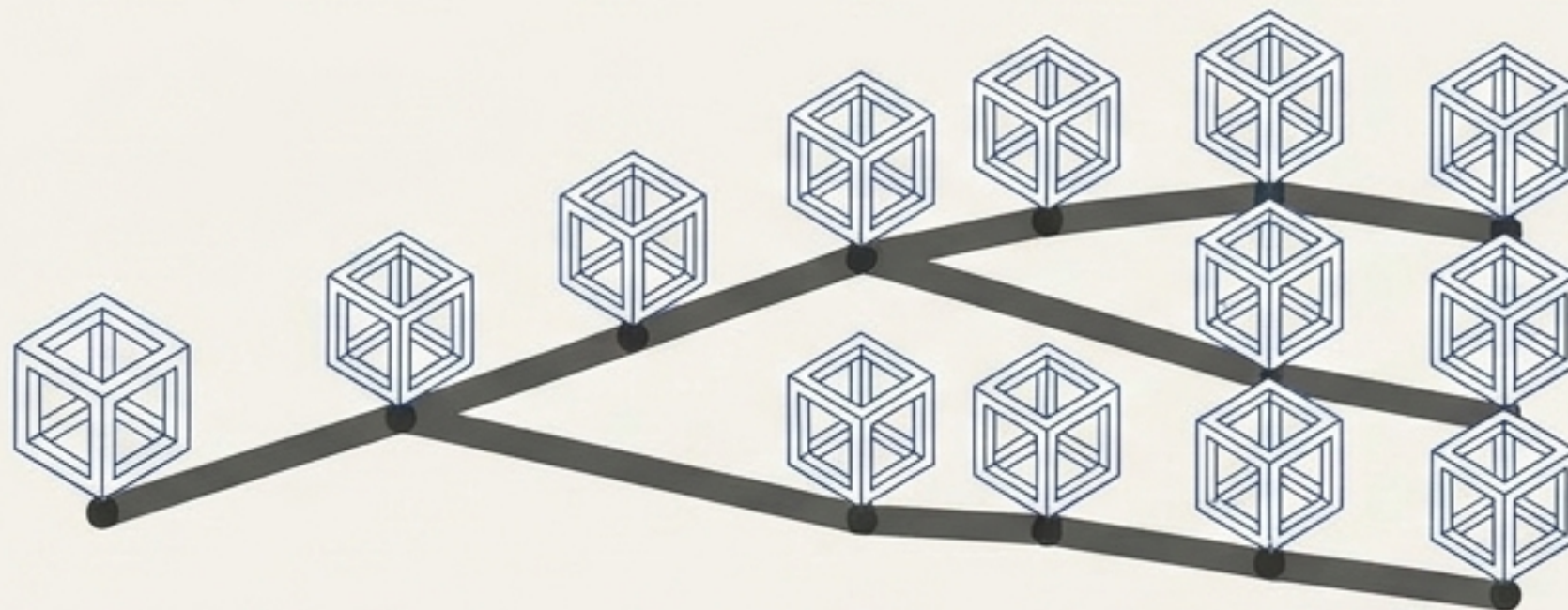


Compose, then Forget. Cross-terms register the relation. Original certainty can be recovered.

'Phase' is not a decorative dial. It is compositional memory. It records distinctions that yield identical visible weights now, but behave differently when combined later.

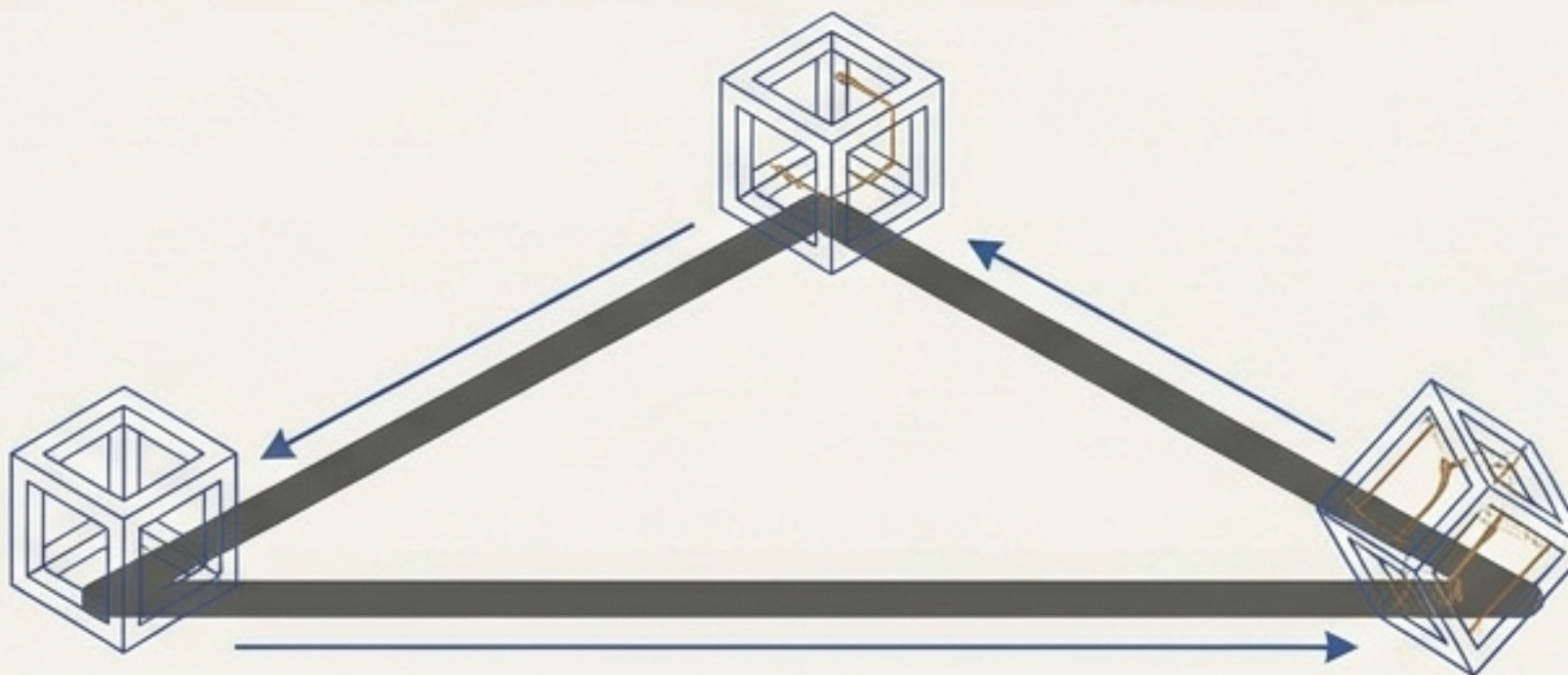
Trees, Cycles, and the Cost of Agreement

The Tree



On a tree, local admissibility is enough. Every new edge explores untouched territory.

The Cycle

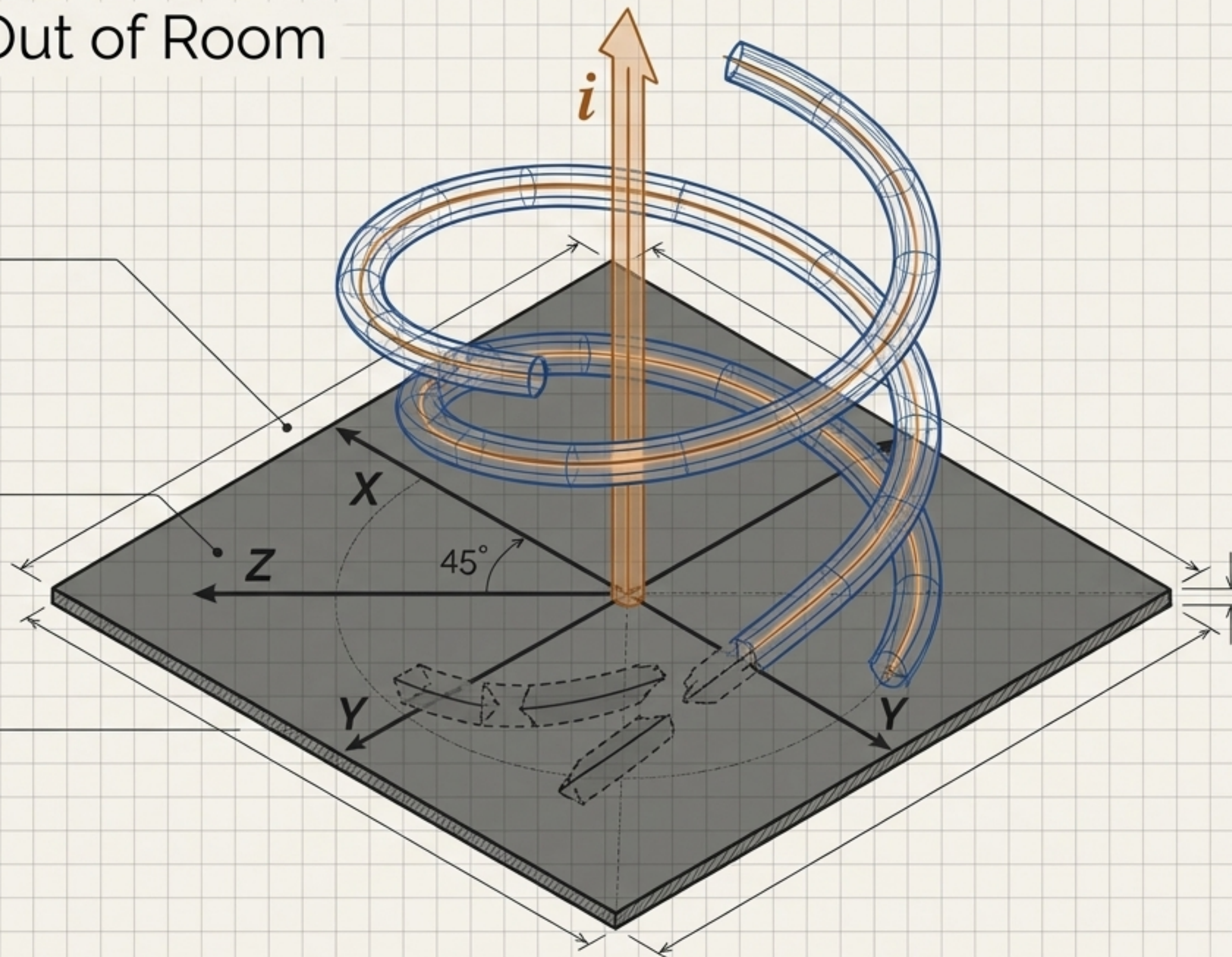


A cycle asks a question no individual edge can answer: after moving through a sequence of perspectives, have we returned to the same frame?

Global coherence requires coordination. Three witnesses can tell perfectly consistent pairwise stories that cannot possibly share a single timeline.

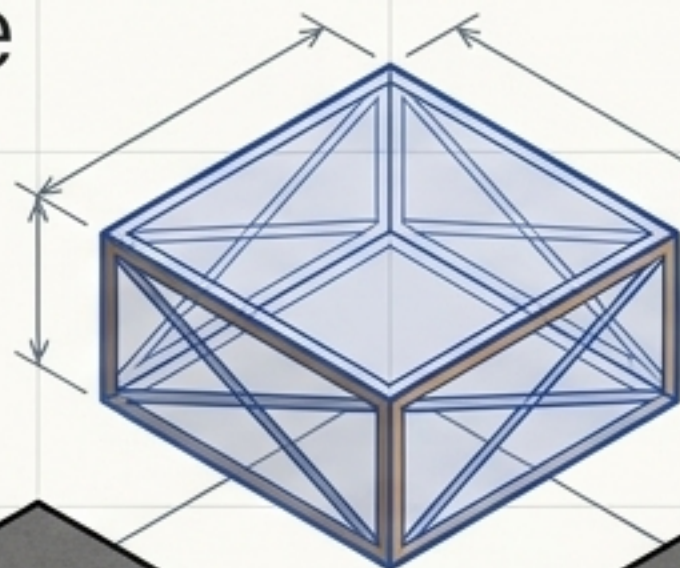
Real Numbers Run Out of Room

1. Any two bases can coexist in a real 2D plane.
2. All three cannot. The final edge hits a finite target set with probability zero.
3. Complex space supplies the missing freedom (phase) to satisfy all constraints at once.

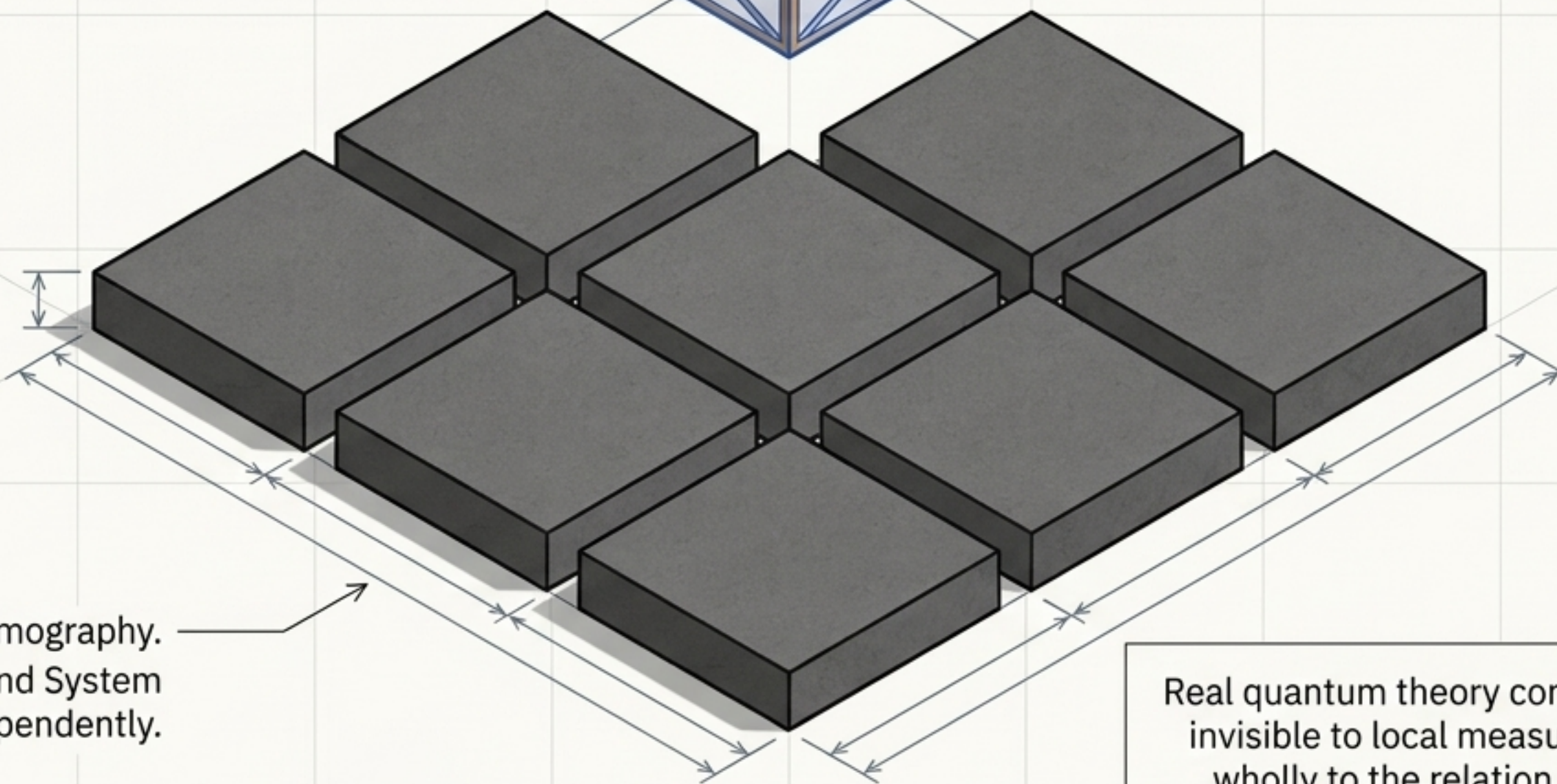


Three mutually unbiased bases (Z, X, Y) form the smallest conceptual cycle.

What the Parts Cannot See



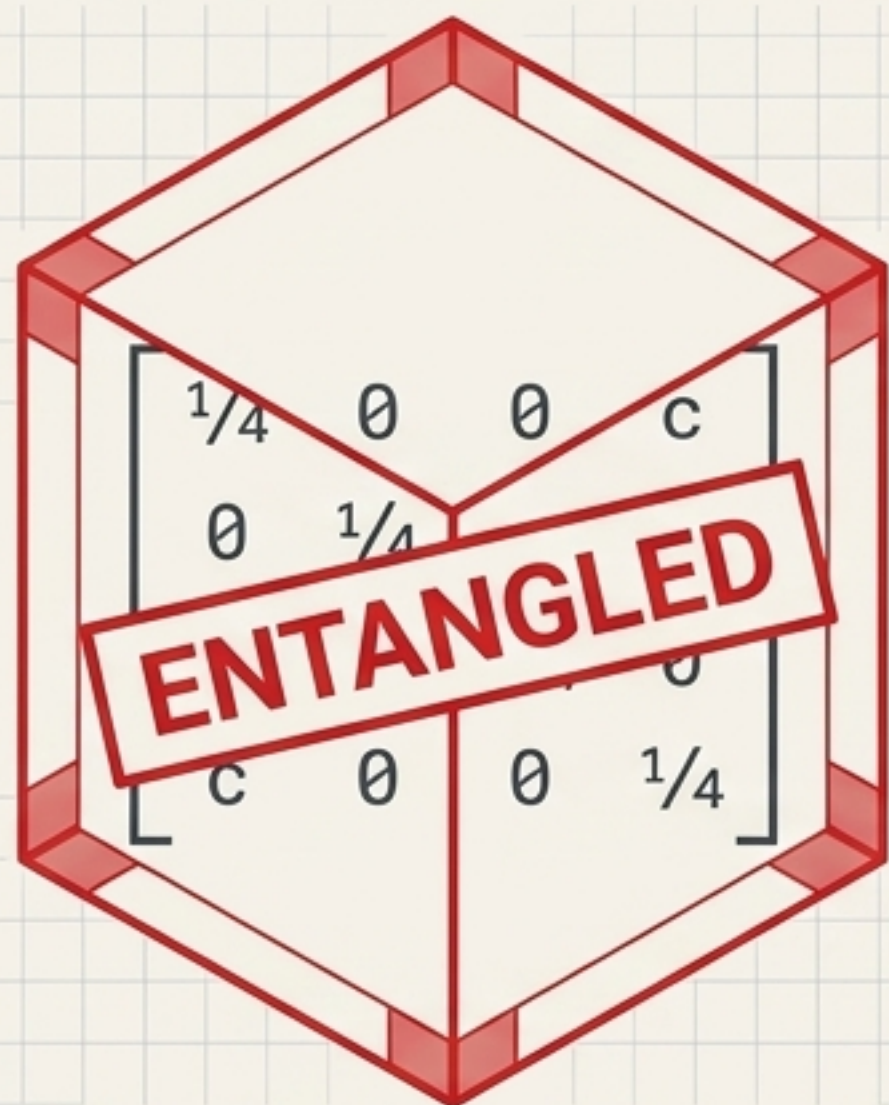
The Antisymmetric Product ($J \otimes J$)
Tracing out either system annihilates
this direction completely.



Local Tomography.
Everything System A and System
B can measure independently.

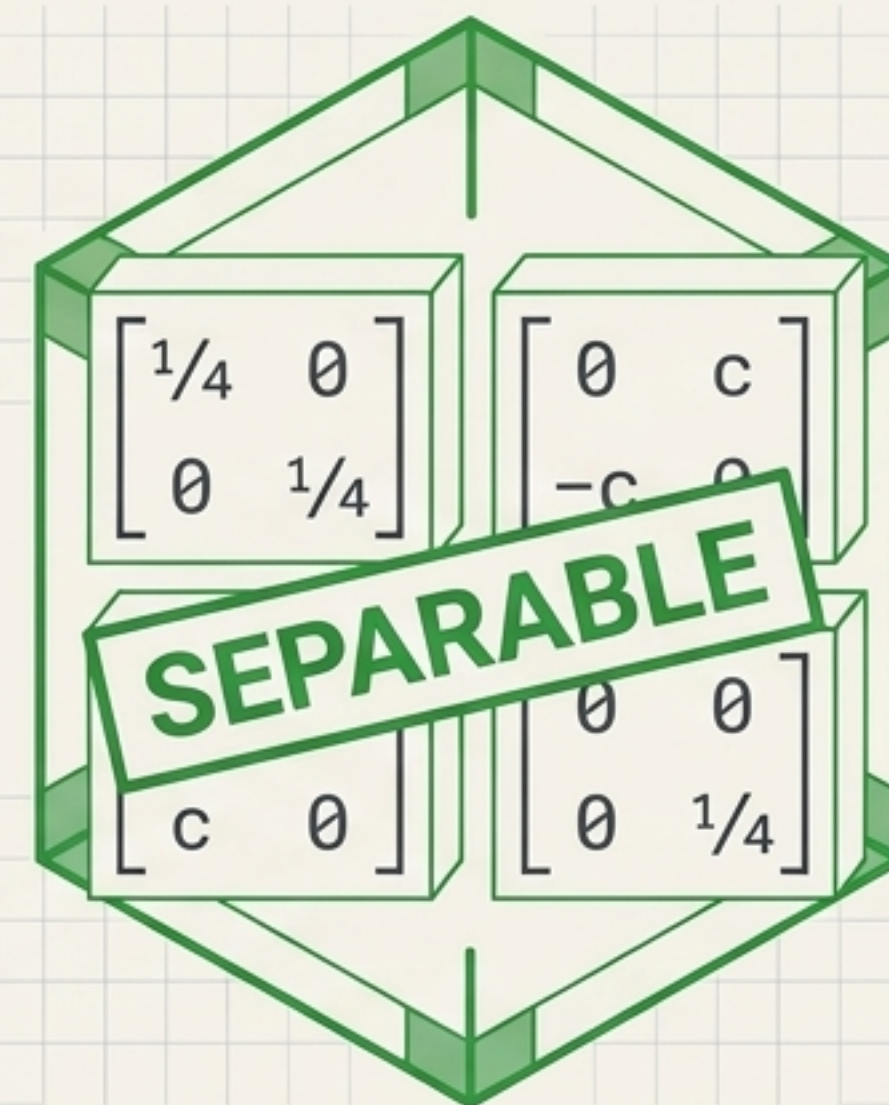
Real quantum theory contains a joint direction
invisible to local measurements. It belongs
wholly to the relation between the parts.

Entanglement as Grammar



Under Real Quantum Theory's rules, local mixtures cannot synthesize the $J \otimes J$ direction. The state is physically inseparable.

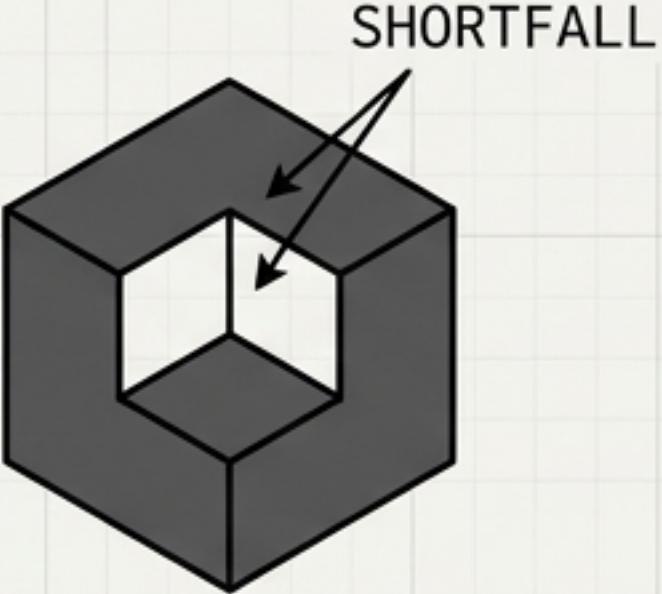
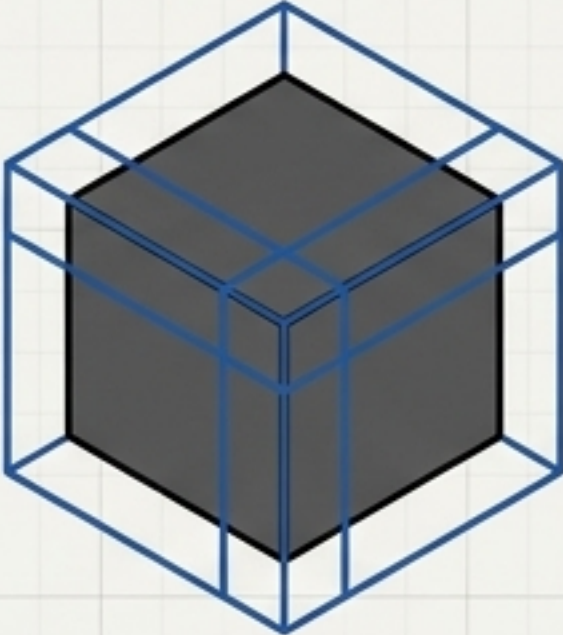
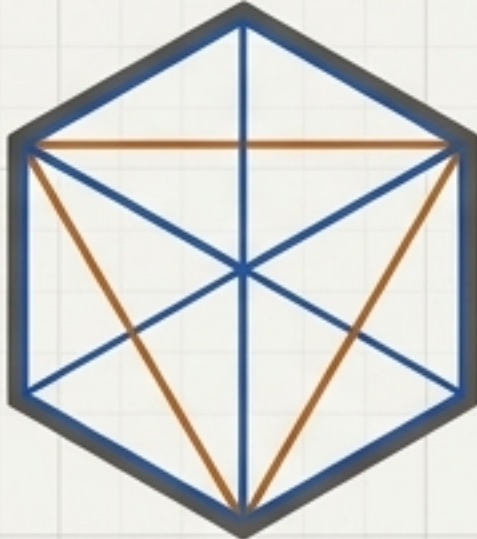
$$\begin{bmatrix} 1/4 & 0 & 0 & c \\ 0 & 1/4 & -c & 0 \\ 0 & -c & 1/4 & 0 \\ c & 0 & 0 & 1/4 \end{bmatrix}$$



Under Complex Quantum Theory, $J \otimes J$ translates to ordinary Pauli matrices. The matrix explicitly decomposes into 4 ordinary product states.

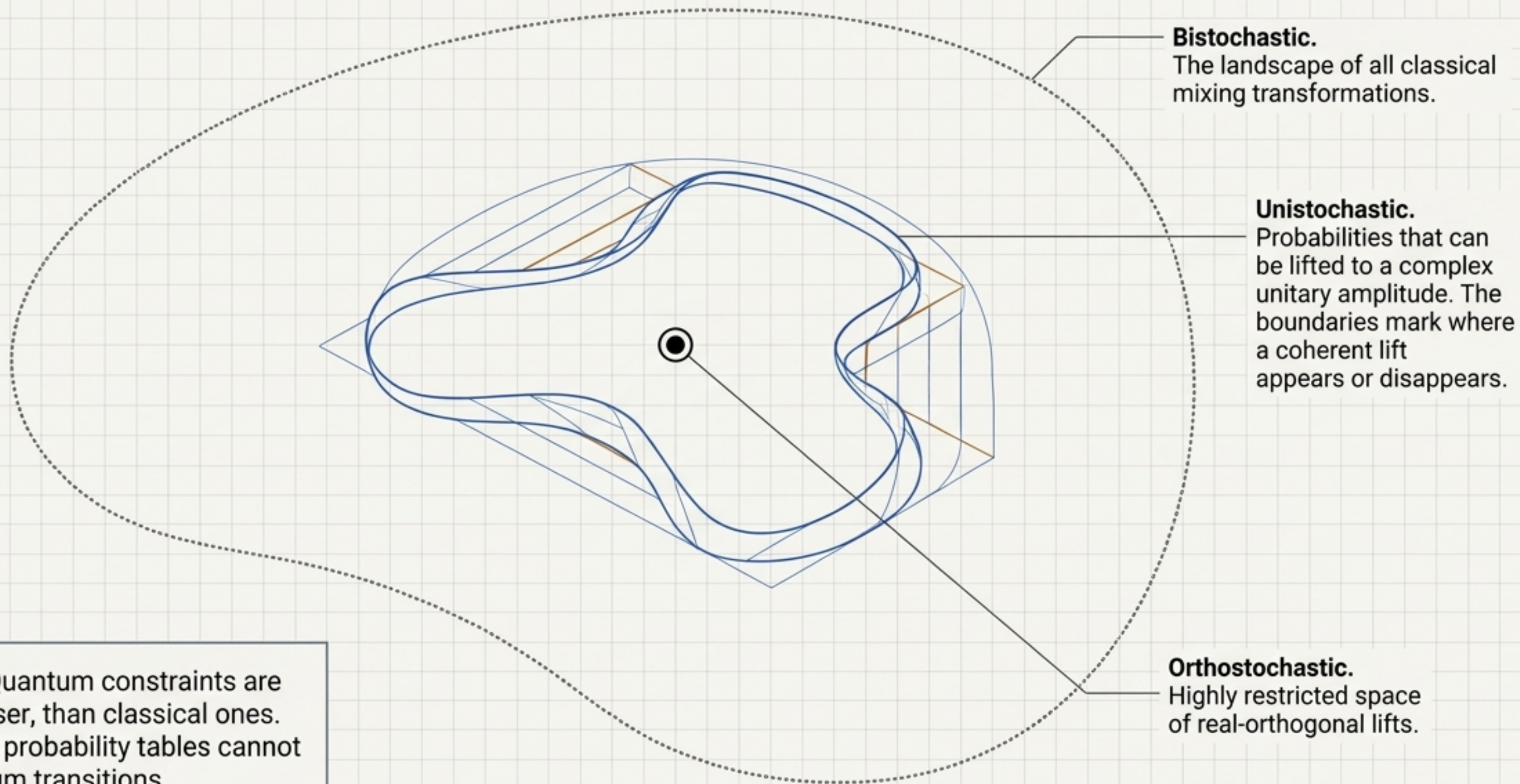
Entanglement is not an intrinsic property of a bare matrix. It is a fact about the fit between a state and the background rules of combination.

The Compositional Fixed Point

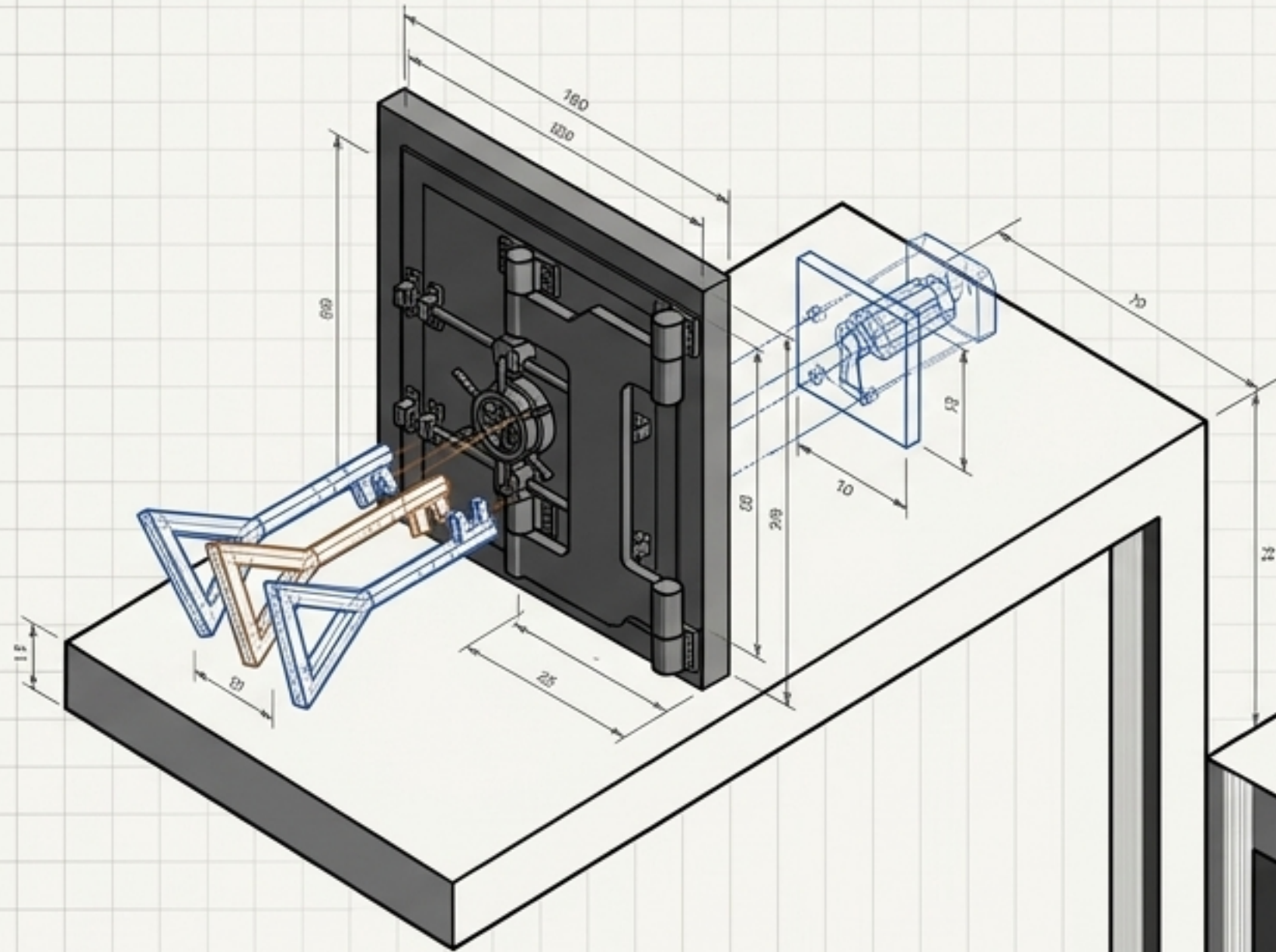
Real Theory	Quaternionic Theory	Complex Theory (The Fixed Point)
• Local parameters come up short. • Contains extra joint directions invisible to local products.  SHORTFALL	• Local parameters overshoot. • Supplies more product parameters than the joint state space can hold.  OVERSHOOT	• Exact fit. • Combines magnitude and orientation so scaling and rotation compose perfectly.  EXACT FIT

“The right question is not “Are complex numbers real?” but “Which invariant work must any empirically adequate representation perform?”

The Geography of Admissibility

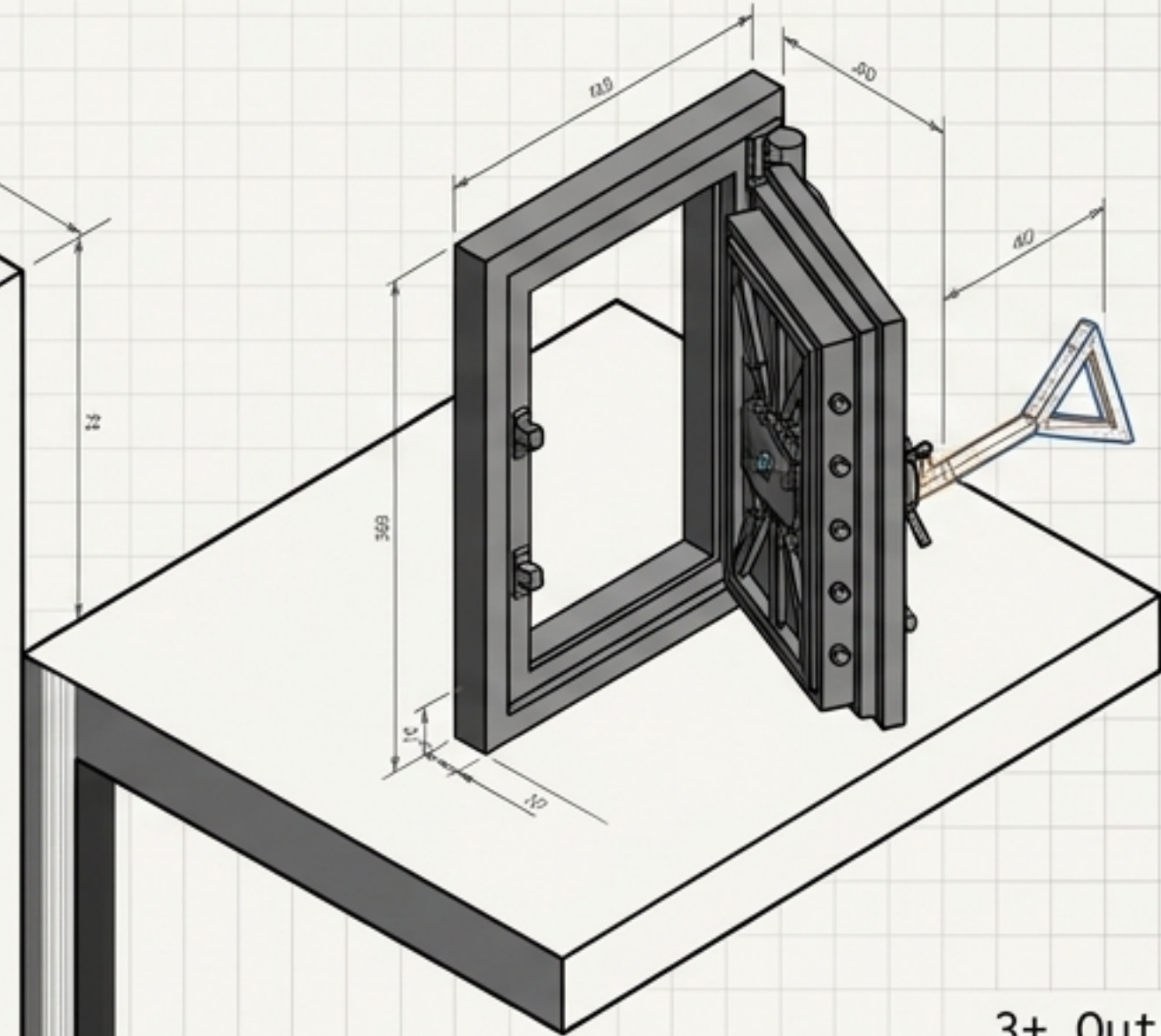


The Cliff at Three Dimensions



2 Outcomes

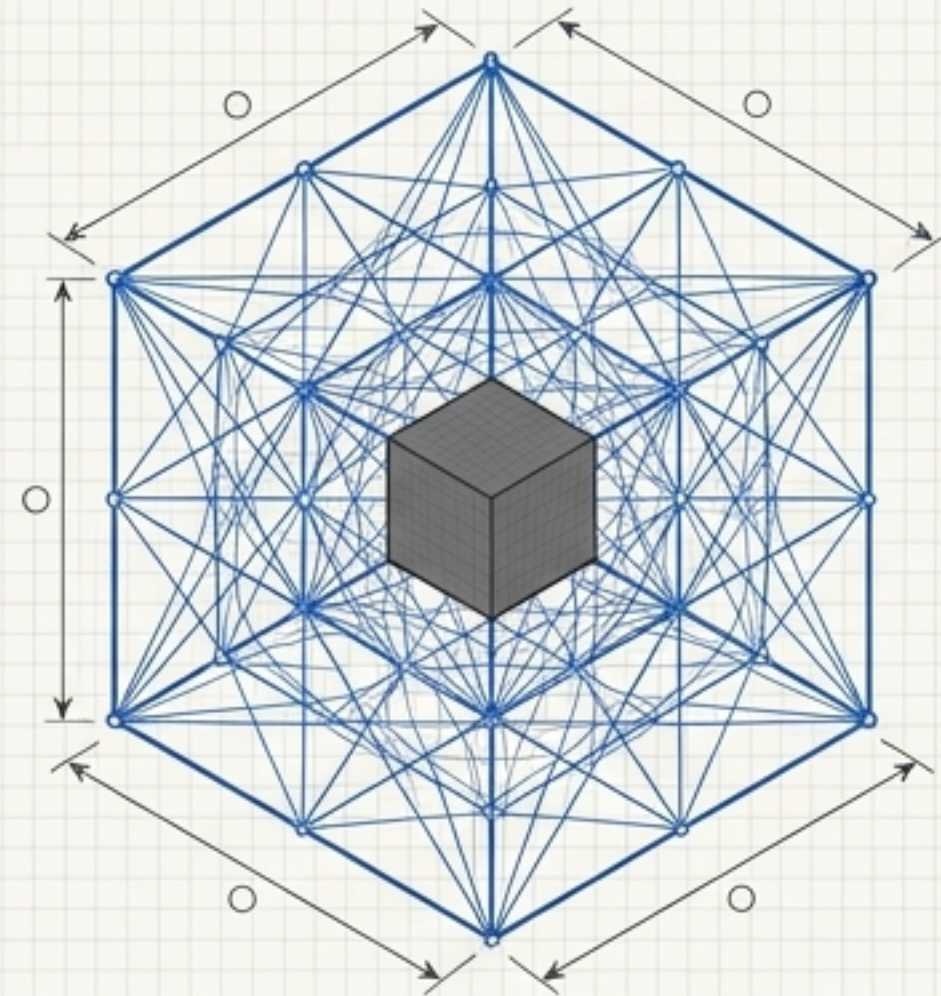
A degenerate coincidence. Every real orthogonal lift is automatically a valid complex lift. We must use closed cycles to detect the difference.



3+ Outcomes

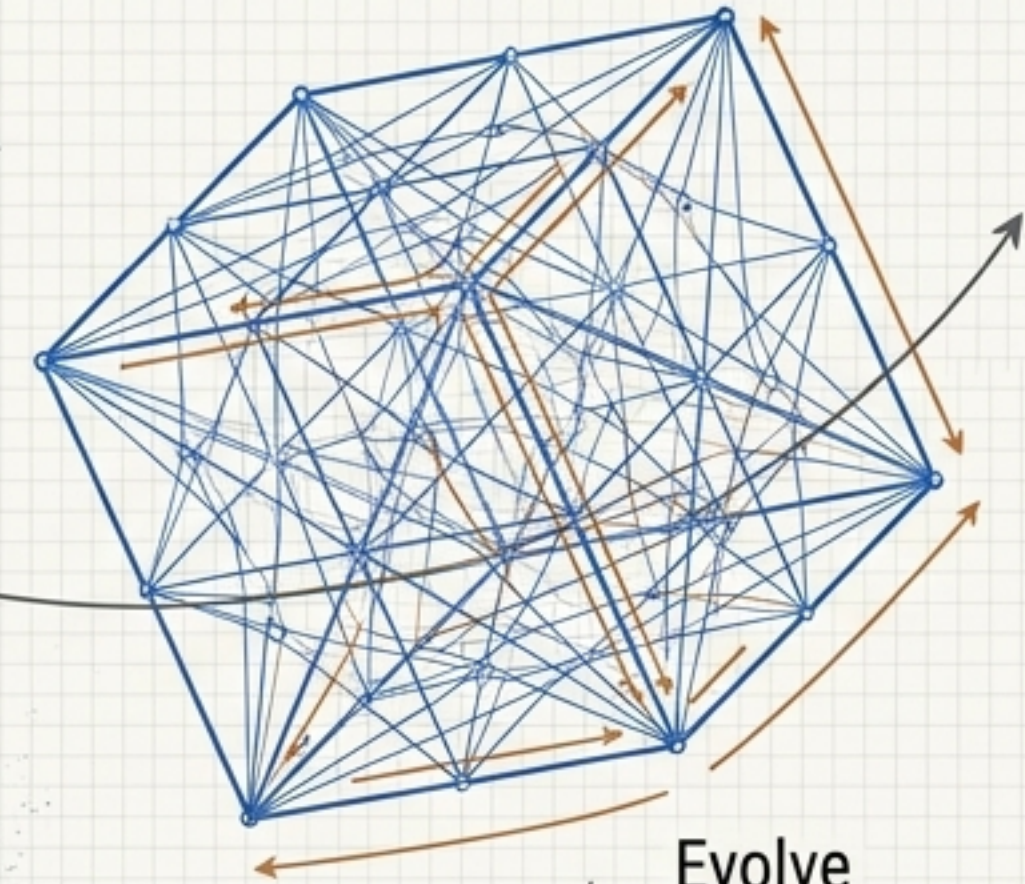
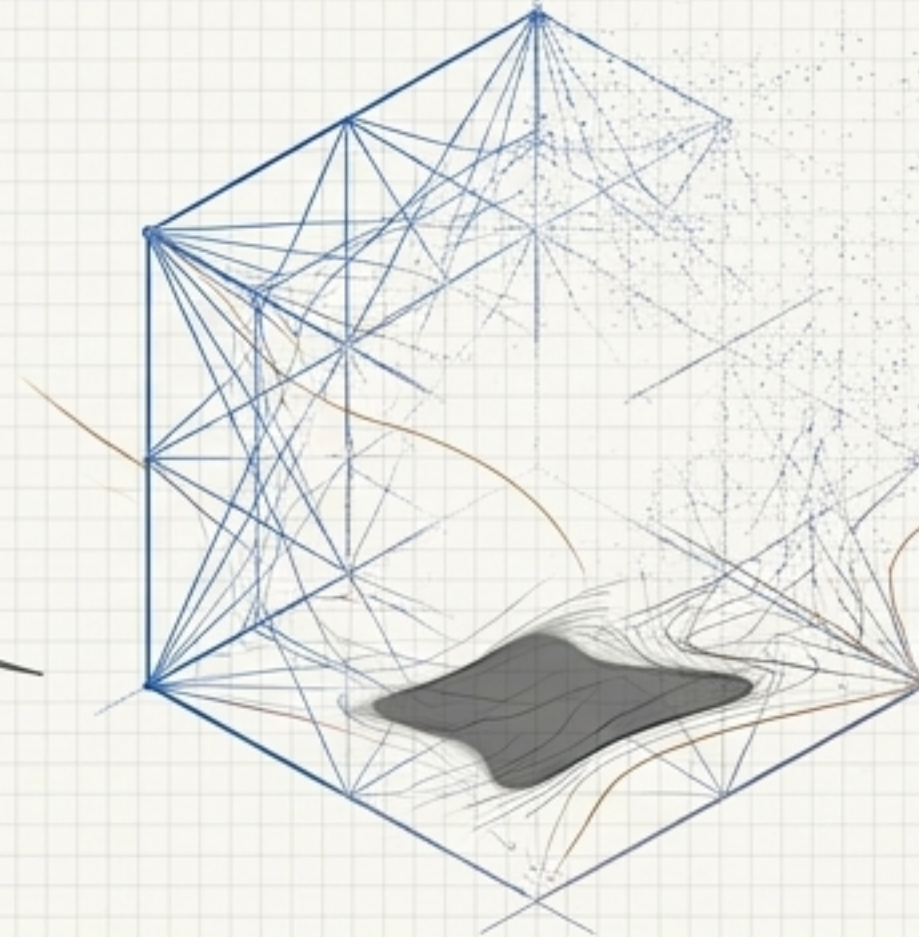
The graph-theoretic complexity collapses. A single 3x3 uniform table refuses a real lift on its own. One badly behaved edge is enough.

How Open Systems Lose Memory



1 Enlarge

Attach an environment. What looks unsharp or noisy locally is perfectly sharp in a larger space (Naimark Dilation).



Evolve

Evolve the joint system reversibly. Phase information flows into vast environmental correlations.

Forget

Trace out the environment. The resulting local shadow experiences decoherence—the loss of accessible phase relations.

The diagram illustrates a mechanical assembly process. It shows a square frame with internal cross-braces. Inside the frame, there are two large gears meshed together. A central shaft passes through the gears. A lever arm is attached to the shaft. The assembly is shown being moved by two robotic grippers from the bottom. Dimensions and arrows indicate movement and assembly sequence.

The Mechanism

The State

The Born Rule is a recovery. By treating a prepared state as a designated vector in a frame, squared amplitudes become state-to-outcome probabilities.

The Dynamics

Schrödinger's equation is not an arbitrary law. It is simply the necessary local differential form of continuous, reversible composition.

The Operator

Hermiticity is not a mystical property. It is merely the mathematical packaging of mutually exclusive sharp outcomes carrying real labels.

The Laboratory

The Foundational Cost Ledger

Framework Commitments (**The Building Code**)

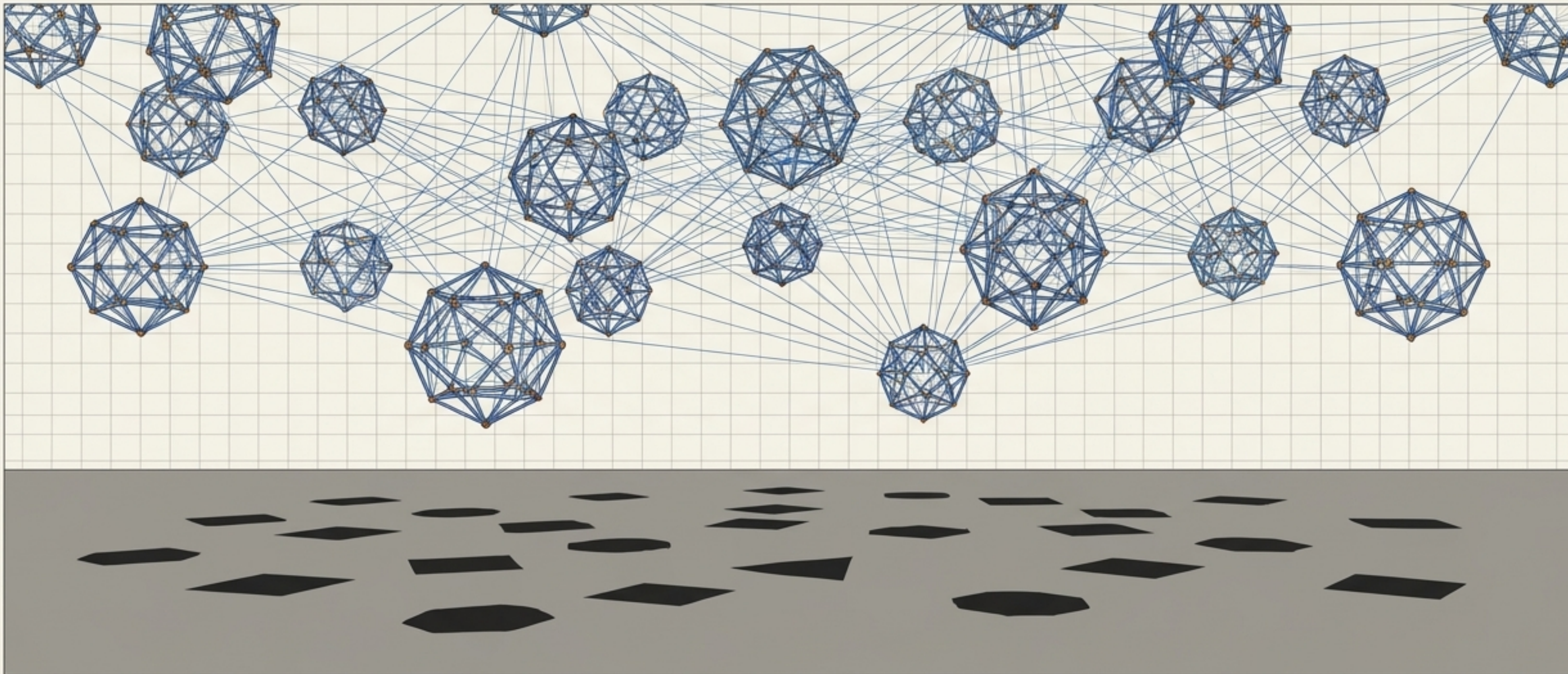
- Finite-dimensional probability spaces
- **Bistochastic** transitions
- **Unistochastic** admissibility
- Convex mixtures
- **Tensor** composition

Operational Bridges (The House Inspection)

- Local tomography
- Pure states as designated completable vectors
- Real-valued measurement labels
- Continuous reversible time evolution

We do not manufacture quantum mechanics from featureless chance.
We choose an admissibility framework, and the mathematics derives the rest.

The World After Probability



Probability tells us the weights.
It does not tell us the orientations.

A record is truthful within its scope,
but local consistency does not guarantee
global consistency.

Quantum mechanics is not a revolt against
reason. It is the most exact demonstration that
continuation reveals what observation forgot.